<u>Module -7: Network fundamental -</u>

4-Explain TCP/IP Networking Mode

➔ 1. Application Layer

- **What it does**: This is where people interact with the network (like when using a web browser, email, or messaging app).
- **Example**: When we type a website URL, this layer helps create the request to load the site.

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2. Transport Layer

- **What it does**: It ensures that the data is delivered correctly and in the right order.
- **Key Protocol**: **TCP** (Transmission Control Protocol) – reliable, and **UDP** (User Datagram Protocol) – faster but less reliable.
- **Example**: If a message gets broken into parts, this layer puts them back in the right order.

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3. Internet Layer

- **What it does**: Handles the delivery of data between computers across networks using **IP (Internet Protocol)**.
- **Key Job**: Finding the best path for the data.
- **Example**: Like writing an address on a letter so it gets to the right house.

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4. Network Access Layer (Link Layer)

- **What it does**: Deals with the physical connection – cables, Wi-Fi, routers, etc.
- **Example**: Like the mail truck or delivery van that carries our letter to the house.

📦 Example (Putting it All Together):

When we send a message on WhatsApp:

- **Application Layer**: we type and hit "Send".
- **Transport Layer**: Breaks the message into packets.
- **Internet Layer**: Decides how to get the packets to your friend.
- **Network Access Layer**: Actually sends the packets through Wi-Fi or mobile data.

6-Explain Operation of Switch

➔ ▯ What is a Switch?

A **switch** is a device used in computer networks to **connect multiple devices** (like computers, printers, and servers) so they can talk to each other.

⚙️ How Does a Switch Work?

1. **Receives Data**: A switch gets data (in little chunks called *frames*) from one device.
2. **Reads the Address**: It looks at the destination **MAC address** (like a device's ID).
3. **Sends to the Right Device**: It forwards the data **only to the device** that needs it—not to all devices.
4. **Learns as It Goes**: The switch remembers which device is on which port, so it gets faster and smarter over time.

🎯 Simple Example:

- we have 4 computers connected to a switch.
- Computer A wants to send a file to Computer C.
- The switch checks where Computer C is connected and sends the file **only to C**, not to A, B, and D.

✅ Why Use a Switch?

- **Faster**: Only sends data to the right device.
- **Efficient**: Reduces unnecessary traffic.
- **Smart**: Learn where devices are over time.